Preliminary report on the emotion classifier project

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Abstract

In this report we share our approach to the facial emotion recognition problem. The main focus of this paper is to give the reader a brief overview over the datasets, the data preprocessing and the model architecture used in our approach.

1. Introduction

Facial Expression Recognition (FER) is pivotal for Human-Computer Interaction, with critical applications spanning biomedical engineering, healthcare, crime detection, and education [6]. While humans interpret these cues intuitively, replicating this capability computationally remains a significant research challenge.

The primary objective of this project is to implement and evaluate Deep Learning architectures trained from scratch to classify six core emotions—anger, disgust, fear, happiness, sadness, and surprise—using constrained 64×64 pixel inputs. Our initial research found out the necessity of a systematic quality filtering strategy, incorporating techniques such as Laplacian blur detection [11], to effectively process noisy benchmarks like RAF-DB. This report documents our methodology, architectural selection, the integration of Explainable AI (XAI) for the final demonstration, and the key insights gained during this preliminary phase.

2. Datasets

To ensure reliable and comparable results, four widely used benchmark datasets from the literature were selected to train and evaluate the proposed models. All datasets contain facial images categorized into the seven target expressions: disgust, fear, happiness, sadness, surprise, and neutral. The datasets employed are Facial Expression Recognition 2013 Plus (FER2013+) [2], Real-World Affective Face Database (RAF-DB) [9], Cohn–Kanade (CK+) [10], and AffectNet [12].

FER2013+ was chosen instead of the original FER2013 due to its improved labelling quality, which has a direct impact on model performance and reliability. In addition, the selected datasets provide diversity in terms of age, ethnicity, and gender, which is essential for improving the generalization capability of the models. Both AffectNet, CK+ and FER+ include an additional contempt label, however, this class is excluded from the experiments, as it falls outside the scope of the present study.

All images are resized to 64×64 pixels to facilitate structural exploration and reduce computational complexity. The data include both grayscale and RGB images. Image formats vary between JPEG and PNG, which does not affect the preprocessing or training pipeline.

Regarding dataset characteristics, the selected subset of AffectNet contains 27,755 RGB images with a resolution of 96×96 pixels in JPEG format. The CK+ dataset consists of 927 grayscale images with a resolution of 48×48 pixels in PNG format. FER2013+ includes 69,263 grayscale images at 112×112 pixels, also in PNG format. Finally, RAF-DB comprises 15,339 images, both grayscale and RGB, with a resolution of 100×100 pixels in JPEG format.

3. Methodology: Data Preprocessing

We prioritize a large dataset, but try to maintain quality. We clean unconstrained images to remove noise that hurts model training.

3.1. Detection and Geometric Filtering

We use MediaPipe for face detection, increasing the size of low-resolution images using **SRGAN** [8] for accuracy, because MediaPipe was trained on bigger images.
Occlusion: Images are removed if no face is found or if key face parts (both eyes, nose) are blocked.
Pose Constraints: We reject images with a Yaw (left/right) rotation over 45 degrees. Pitch (up/down) and Roll (tilt) are also limited to avoid extreme angles, ensuring features are visible.

3.2. Backgrounds and Image Quality

Cropping: If the face is small ($< 10\%$ of the image), we crop to center it. We keep some background to help the model handle environmental noise.
Channel Standardization: We convert any single-channel grayscale images to 3-channel mode to ensure dimensional

consistency with the model input, though the visual appearance remains unchanged.
Blur & Deduplication: We remove blurry images using the **Laplacian method** [1], this will also remove cartoon images and we remove exact duplicates using **SHA-256 hashing**, or **Perceptual Hashing** [5] if more performance is needed.

3.3. Imbalance and Reduction of Background Noise

To reduce the effect of background noise, we will experiment using an **Attention-Hybrid CNN**. This architecture learns to focus on facial features over background noise. We handle the imbalance in emotion classes using **Class-Balanced Focal Loss** [3] instead of reducing the data, further more about it.

4. Model Architectures

To address the facial emotion recognition task (FER), our group investigates multiple convolutional neural networks (CNNs) that have demonstrated strong performance in image classification and facial analysis tasks. CNNs build the base for most of our models. This decision is based on according literature emphasizing the clear performance advantage of CNNs compared to other approaches such as traditional machine learning models [4]. Each team member focuses on implementing and evaluating selected architectures in order to identify the most suitable model for achieving a high classification accuracy while maintaining robust generalization performance.

- Dmytro Dumanskyy:
- Tiago Kienzle Hagen: ResNet, CERN, Poster++
- Aleksandar Neshov: ResNet, EfficientNet, DenseNet
- Daniel Pfefferkorn: GoogLeNet, VGG-19, EmoNeXt

Maximizing the accuracy of the FER task represents the core goal of our group. Apart from the aforementioned aspects, the actual architecture of our neural network plays a key role in achieving this goal to the best of our ability. In the follwoing section we want to highlight our decision making process.

4.1. Criteria for model selection

The suitability of a model for this project is evaluated based on the following criteria:

Firstly the design should be well documented in the literature to ensure scientifically proven effectiveness. For a model to be taken into closer consideration it should have reported strong results on standart benchmarks.

Secondly the architecture should be capable of extractig important facial featrures such as the eyes, the eyebrows and the mouth meaningfully.

Finally models that are prone to overfitting are seen as unfavorable for our project. The chosen network should be able to learn distinctive features well while not overly relying on a high number of samples to achieve acceptable results. Since FER datasets are limited in sample size, we will prioritize models that shine through their data efficiency.

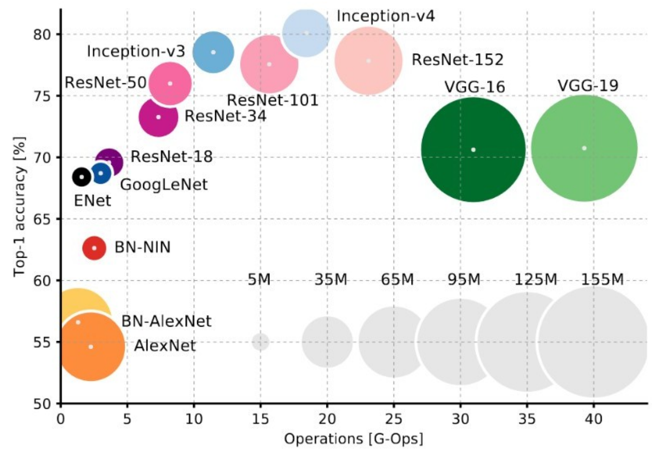

Figure 1. This figure from Kopalidis et al. [6] was used as a general reference as to which models to take into consideration.

4.2. Excluded model architectures

Not all types of models were categorized as suitable for the task at hand. Vision Transformers (ViT) are highly data hungry and tyically rely on large-scale pretraining, making them suboptimal for training from scratch on limited FER datasets. Furthermore any outdated neural networks that are no longer competitive in terms of accuracy or efficiency were used as baselines for comparison.

4.3. Training setup

We decided to research common training parameters used in scientific papers that yielded good results. All models are trained from scratch using a consistent training protocol to ensure fair comparison.

We decided to use Cross-Entropy loss as our loss function since it's widely used in similar classification tasks. Using Stochastic gradient descent (SGD) as our optimizer we are going to train the networks for approximately 30-50 epochs depending on individual model convergence. Furthermore we plan on exploring attention mechanisms to improve robustness against facial occlusions.

To ensure a consistent baseline fo rmodel evaluation all models are trained and tested on a predefined dataset.

4.4. Model selection strategy

We are currently in theprocess of deciding on a final neural network design for the entire group. Each team member is implementing their own model based on the architectures described above. The final model will be decided through the highest test accuracy on our collective test set.

5. Explainable AI

In order to portray the model's regions of interest in the image we plan on using GradCAM, which is a common method used for visualizing and interpreting networks by highlighting the image regions that contribute most strongly to a model's prediction.

6. Demo Implementation

To demonstrate the practical application of our work, we will implement a real-time video pipeline using **OpenCV**, adapting the approach described by Kumar [7].

The planned system consists of two main steps:

1. **Face Detection:** We will use OpenCV's Haar cascade classifiers to scan the video frame. This algorithm identifies the coordinates of the face, allowing us to isolate the facial region from the background.
2. **Classification:** The detected face region will be cropped and resized to 64×64 pixels. This processed image is then passed to our model to classify the emotion.

The final output will display the video feed with a bounding box and the emotion label overlaid in real-time.

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